

Vertigo

Overview

Vertigo is a neurological symptom characterized by the false sensation that a person or their surroundings are spinning or moving, even when there is no actual movement. It is not a disease itself but rather a symptom of disorders affecting the inner ear, vestibular nerve, or brain. Vertigo can significantly interfere with daily activities by affecting balance, coordination, and spatial orientation.

Episodes of vertigo may last for a few seconds, several minutes, hours, or even days depending on the underlying cause. Some individuals experience occasional attacks, while others may develop recurrent episodes. Although most cases are caused by problems in the inner ear, vertigo can sometimes indicate a serious neurological condition such as a stroke or brain tumor.

Early diagnosis is important because treatment depends on identifying the underlying cause.

Types of Vertigo

Vertigo is broadly classified into two major categories.

1. Peripheral Vertigo

Peripheral vertigo is the most common type and results from disorders affecting the inner ear or vestibular nerve.

Common causes include:

- Benign Paroxysmal Positional Vertigo (BPPV)
- Vestibular neuritis
- Labyrinthitis
- Ménière's disease

Symptoms are often severe but usually improve with appropriate treatment.

2. Central Vertigo

Central vertigo occurs due to disorders affecting the brain, particularly the brainstem or cerebellum.

Common causes include:

- Stroke
- Multiple sclerosis
- Brain tumor
- Migraine
- Head injury

Central vertigo is generally less common but may indicate a serious neurological emergency requiring immediate medical evaluation.

Causes

Vertigo can result from several medical conditions affecting balance mechanisms.

Common causes include:

- Benign Paroxysmal Positional Vertigo (BPPV)
- Vestibular neuritis
- Labyrinthitis
- Ménière's disease
- Migraine
- Stroke
- Multiple sclerosis
- Brain tumors
- Head trauma
- Certain medications

The most common cause is **Benign Paroxysmal Positional Vertigo (BPPV)**, in which tiny calcium crystals inside the inner ear become displaced.

Risk Factors

Several factors increase the likelihood of developing vertigo.

These include:

- Increasing age
- Previous episodes of vertigo
- Head injury
- Inner ear infections
- Migraine
- High blood pressure
- Diabetes
- Stroke
- Certain medications
- Family history of vestibular disorders

Managing underlying health conditions may reduce the risk of recurrent vertigo.

Symptoms

Symptoms vary depending on the underlying cause.

Common symptoms include:

- Spinning sensation
- Feeling that the room is spinning
- Loss of balance
- Dizziness
- Difficulty walking
- Nausea
- Vomiting
- Unsteady gait

- Motion sensitivity
- Sweating
- Ringing in the ears (tinnitus)
- Hearing loss (in some conditions)
- Blurred vision
- Difficulty focusing

Symptoms are often worsened by sudden head movements.

Benign Paroxysmal Positional Vertigo (BPPV)

BPPV is the most common cause of peripheral vertigo.

Characteristics include:

- Brief episodes of spinning
- Triggered by changes in head position
- Rolling over in bed
- Looking upward
- Bending forward
- Symptoms lasting less than one minute

BPPV occurs when tiny calcium crystals move into the semicircular canals of the inner ear.

Ménière's Disease

Ménière's disease is a chronic inner ear disorder.

Symptoms include:

- Recurrent vertigo attacks
- Hearing loss
- Ringing in the ears
- Feeling of fullness in the ear

Episodes may last from 20 minutes to several hours.

Diagnosis

Diagnosis involves identifying the underlying cause of vertigo.

Healthcare providers may perform:

- Detailed medical history
- Neurological examination
- Ear examination
- Balance assessment
- Dix-Hallpike maneuver
- Hearing tests
- MRI of the brain
- CT scan
- Blood tests (when appropriate)

The Dix-Hallpike maneuver is commonly used to diagnose BPPV by reproducing vertigo symptoms through specific head movements.

Treatment

Treatment depends on the underlying cause.

Medications

Common medications include:

- Meclizine
- Betahistine
- Antiemetic medications
- Vestibular suppressants

These medications help relieve dizziness and nausea but may not treat the underlying cause.

Repositioning Maneuvers

Patients with BPPV often benefit from:

- Epley maneuver
- Semont maneuver

These procedures reposition displaced calcium crystals within the inner ear.

Vestibular Rehabilitation

Vestibular rehabilitation therapy includes exercises that help the brain adapt to balance disturbances.

Exercises focus on:

- Balance improvement
 - Eye movement control
 - Walking stability
 - Coordination
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Treatment of Underlying Conditions

Specific treatment depends on the diagnosis.

Examples include:

- Antibiotics for certain infections
 - Corticosteroids for vestibular neuritis
 - Stroke management
 - Migraine treatment
 - Surgery in selected cases of Ménière's disease
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Possible Complications

Untreated vertigo may lead to:

- Falls
- Injuries
- Difficulty walking
- Anxiety
- Reduced confidence
- Motion sickness
- Difficulty performing daily activities
- Reduced quality of life

Repeated falls are particularly concerning in older adults.

Lifestyle Management

Patients can reduce vertigo symptoms by:

- Moving slowly when changing position
- Staying hydrated
- Avoiding sudden head movements
- Performing prescribed vestibular exercises
- Using support while walking if necessary
- Avoiding driving during severe episodes
- Managing stress
- Limiting alcohol consumption
- Getting adequate sleep

Patients with Ménière's disease may also be advised to reduce salt intake.

When to Seek Medical Attention

Immediate medical evaluation is recommended if vertigo is associated with:

- Sudden weakness
- Facial drooping
- Slurred speech
- Double vision
- Severe headache
- Loss of consciousness
- Persistent vomiting
- Difficulty walking
- New hearing loss
- Chest pain
- Seizures

These symptoms may indicate a stroke or another serious neurological condition requiring emergency treatment.

Patient Advice

Patients experiencing recurrent vertigo should seek medical evaluation rather than self-treating with over-the-counter medications. Following prescribed vestibular exercises, attending follow-up appointments, and identifying triggers can help reduce future episodes. Patients should avoid activities such as driving or climbing heights during active vertigo attacks to reduce the risk of injury.

Summary

Vertigo is a symptom characterized by the false sensation of spinning or movement, usually resulting from disorders of the inner ear or the central nervous system. The most common cause is Benign Paroxysmal Positional Vertigo (BPPV), although conditions such as vestibular neuritis, Ménière's disease, stroke, migraine, and brain tumors can also cause vertigo. Diagnosis involves neurological examination, balance testing, and imaging when necessary. Treatment includes medications, repositioning maneuvers, vestibular rehabilitation, and management of the underlying condition. Early diagnosis improves recovery and helps prevent complications such as falls and injuries.

